WASP-2b

R-1261

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_200614` · created 2026-07-31T12:49:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459014.7947 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.047
Transit depth (Rp/Rs)^2	0.99 +/- 0.93 [%]
Orbital Inclination (inc)	82.34 +/- 2.87
Airmass coefficient 1 (a1)	1.01 +/- 0.54
Airmass coefficient 2 (a2)	0.01 +/- 0.23
Scatter in the residuals of the lightcurve fit is	54.97 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.51
Transit Duration (day)	0.031 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.099 ± 0.047 vs 0.1326 (deviation 0.71σ)
Duration fit vs published	0.031 d vs 0.0737 d (deviation 1.26σ)
Mid-time O–C	-9.7 min
Depth SNR	2.1
Residual scatter	54.97 %

Light curve

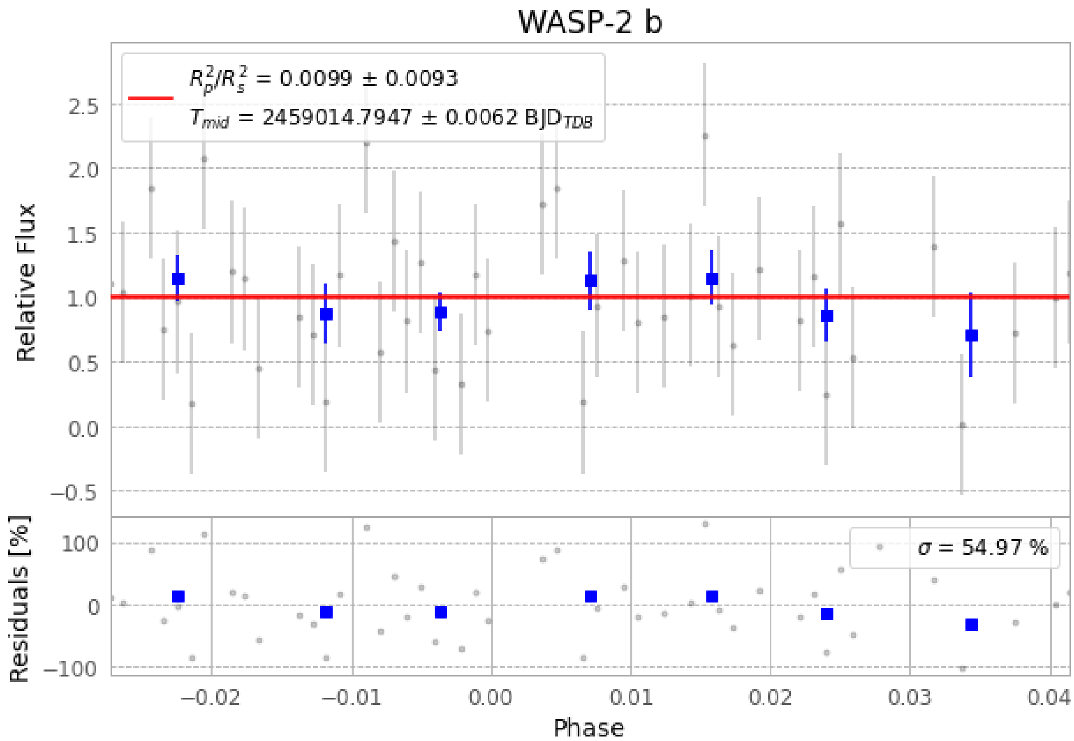

Figure 1 — FinalLightCurve_WASP-2 b_2020-06-13.png (SHA-256 2e38a123421b5d99…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [101, 326], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 79bb174226b28c00…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

72 FITS frames (47 MB), `WASP-2200614053012.FITS` → `WASP-2200614090612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-200614.log (7868e51680c4…), FinalLightCurve_WASP-2 b_2020-06-13.png (2e38a123421b…), FinalLightCurve_WASP-2 b_2020-06-13.csv (c5e16e0898d9…)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)